CHE 109: Engineering Chemistry – I
Summer-2025
Properties of gases
Chapter 6

Prof. Dr. Md. Kamal Hossain
Adjunct Faculty
East West University

5.3 The Gas Laws

The gas laws we will study in this chapter are the product of countless experiments on the physical properties of gases that were carried out over several centuries. Each of these generalizations regarding the macroscopic behavior of gaseous substances represents a milestone in the history of science. Together they have played a major role in the development of many ideas in chemistry.

The Pressure-Volume Relationship: Boyle's Law

In the seventeenth century, Robert Boyle[†] studied the behavior of gases systematically and quantitatively. In one series of studies, Boyle investigated the pressure-volume relationship of a gas sample. Typical data collected by Boyle are shown in Table 5.2. Note that as the pressure (P) is increased at constant temperature, the volume (V) occupied by a given amount of gas decreases. Compare the first data point with a pressure of 724 mmHg and a volume of 1.50 (in arbitrary unit) to the last data point with a pressure of 2250 mmHg and a volume of 0.58. Clearly there is an inverse relationship between pressure and volume of a gas at constant temperature. As the pressure is increased, the volume occupied by the gas decreases. Conversely, if the applied pressure is decreased, the volume the gas occupies increases. This relationship is now known as ***Boyle's law,*** which states that *the pressure of a fixed amount of gas at a constant temperature is inversely proportional to the volume of the gas.*

The apparatus used by Boyle in this experiment was very simple (Figure 5.5). In Figure 5.5(a), the pressure exerted on the gas is equal to atmospheric pressure and the volume of the gas is 100 mL. (Note that the tube is open at the top and is therefore exposed to atmospheric pressure.) In Figure 5.5(b), more mercury has been added to double the pressure on the gas, and the gas volume decreases to 50 mL. Tripling the pressure on the gas decreases its volume to a third of the original value [Figure 5.5(c)].

The pressure applied to a gas is equal to the gas pressure.

We can write a mathematical expression showing the inverse relationship between pressure and volume:

$$P \propto \frac{1}{V}$$

where the symbol $\propto$ means *proportional to*. We can change $\propto$ to an equals sign and write

$$P = k_1 \times \frac{1}{V} \tag{5.1a}$$

where k_1 is a constant called the *proportionality constant*. Equation (5.1a) is the mathematical expression of Boyle's law. We can rearrange Equation (5.1a) and obtain

$$PV = k_1 \tag{5.1b}$$

This form of Boyle's law says that the product of the pressure and volume of a gas at constant temperature and amount of gas is a constant. The top diagram in Figure 5.6 is a schematic representation of Boyle's law. The quantity n is the number of moles of the gas and R is a constant, to be defined in Section 5.4. We will see there that the proportionality constant k_1 in Equation (5.1) is equal to nRT.

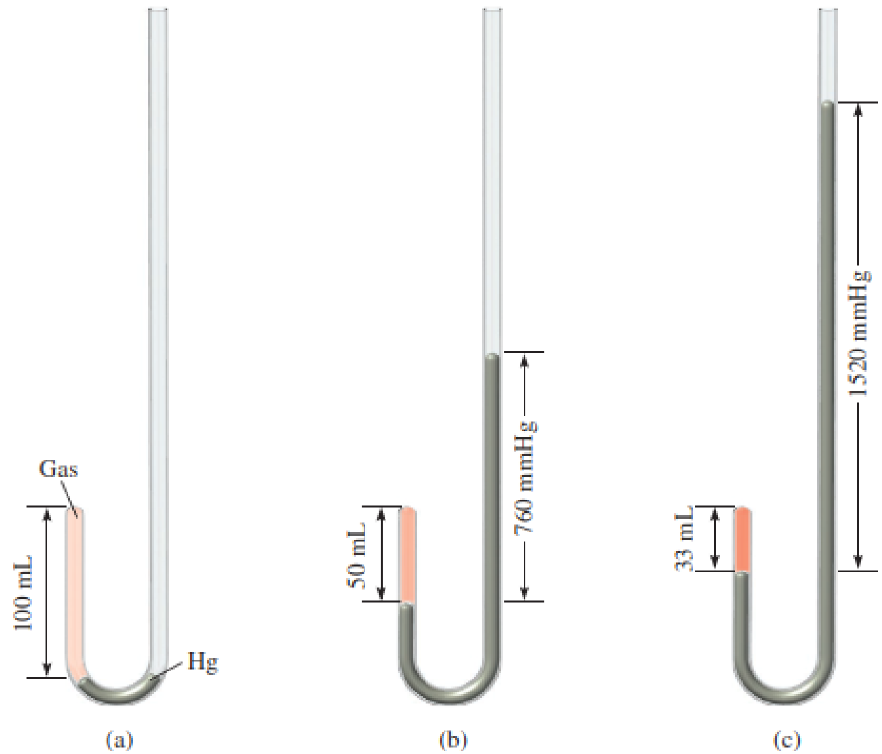


Figure 5.5 *Apparatus for studying the relationship between pressure and volume of a gas. (a) The levels of mercury are equal and the pressure of the gas is equal to the atmospheric pressure (760 mmHg). The gas volume is 100 mL. (b) Doubling the pressure by adding more mercury reduces the gas volume to 50 mL. (c) Tripling the pressure decreases the gas volume to one-third of the original value. The temperature and amount of gas are kept constant.*

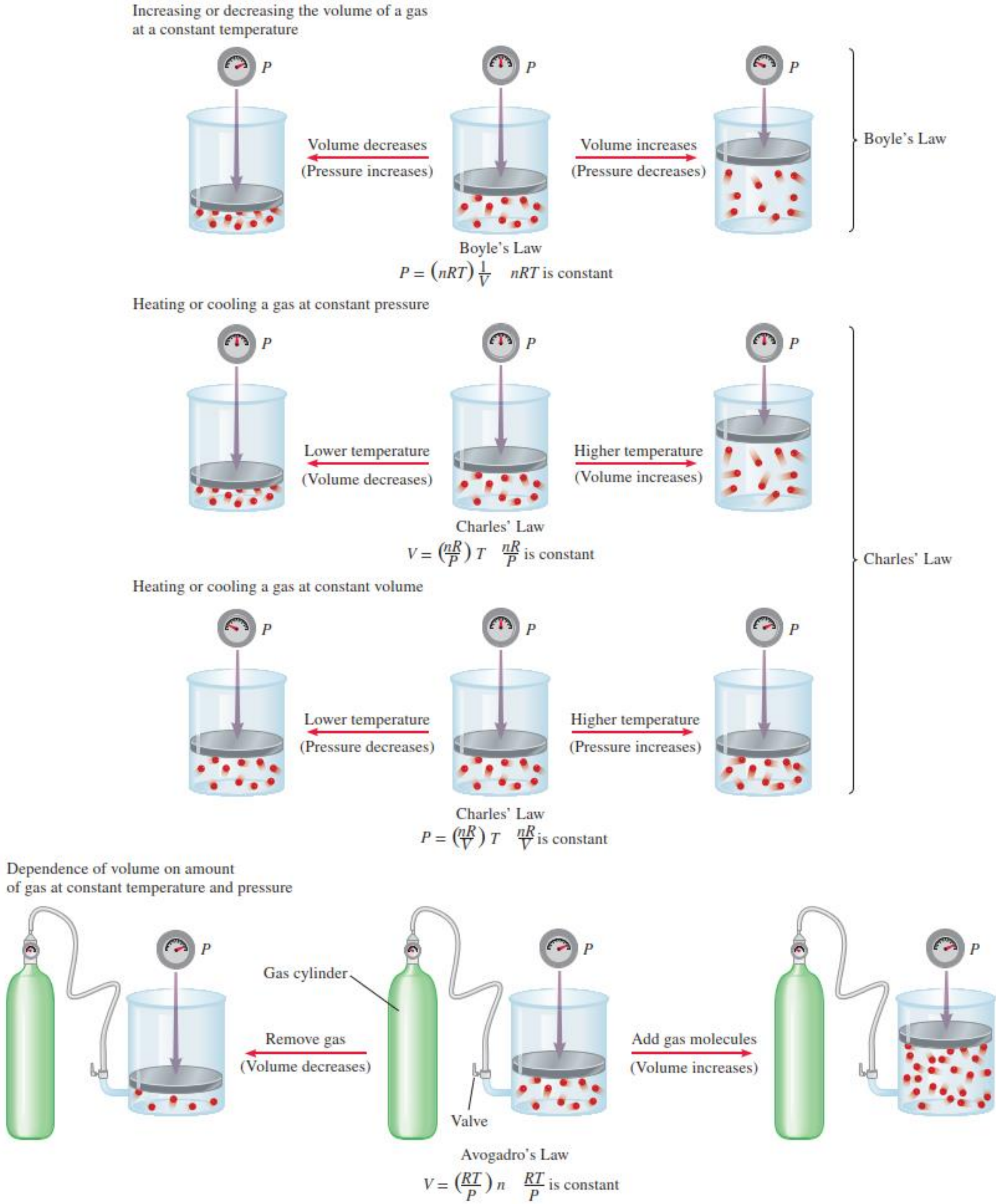


Figure 5.6 *Schematic illustrations of Boyle's law, Charles' law, and Avogadro's law.*

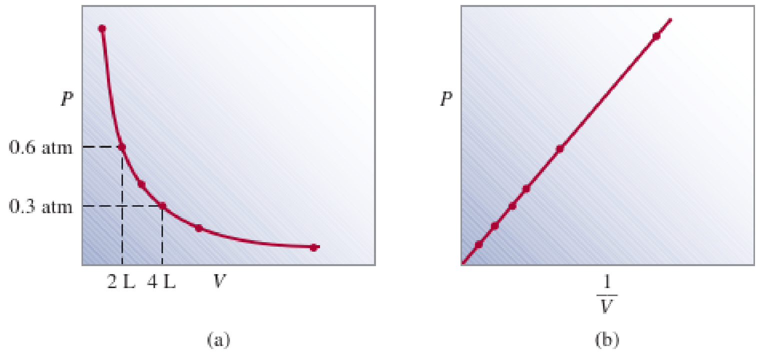


Figure 5.7 *Graphs showing variation of the volume of a gas with the pressure exerted on the gas, at constant temperature. (a) P versus V. Note that the volume of the gas doubles as the pressure is halved. (b) P versus 1/V. The slope of the line is equal to k_1.*

Figure 5.7 shows two conventional ways of expressing Boyle's findings graphically. Figure 5.7(a) is a graph of the equation $PV = k_1$; Figure 5.7(b) is a graph of the equivalent equation $P = k_1 \times 1/V$. Note that the latter is a linear equation of the form $y = mx + b$, where $b = 0$ and $m = k_1$.

Although the individual values of pressure and volume can vary greatly for a given sample of gas, as long as the temperature is held constant and the amount of the gas does not change, P times V is always equal to the same constant. Therefore, for a given sample of gas under two different sets of conditions at constant temperature, we have

$$P_1V_1 = k_1 = P_2V_2$$

or

$$P_1V_1 = P_2V_2 \tag{5.2}$$

where V_1 and V_2 are the volumes at pressures P_1 and P_2, respectively.

The Temperature-Volume Relationship: Charles' and Gay-Lussac's Law

Boyle's law depends on the temperature of the system remaining constant. But suppose the temperature changes: How does a change in temperature affect the volume and pressure of a gas? Let us first look at the effect of temperature on the volume of a gas. The earliest investigators of this relationship were French scientists, Jacques Charles[†] and Joseph Gay-Lussac.[‡] Their studies showed that, at constant pressure, the volume of a gas sample increases when heated and decreases when cooled (Figure 5.8). The quantitative relations involved in changes in gas temperature and volume turn out to be remarkably consistent. For example, we observe an interesting phenomenon when we study the temperature-volume relationship at various pressures. At any given pressure, the plot of volume versus temperature yields a straight line. By extending the line to zero volume, we find the intercept on the temperature axis to be −273.15°C. At any other pressure, we obtain a different straight line for the volume-temperature plot, but we get the *same* zero-volume temperature intercept at −273.15°C (Figure 5.9). (In practice, we can measure the volume of a gas over only a limited temperature range, because all gases condense at low temperatures to form liquids.)

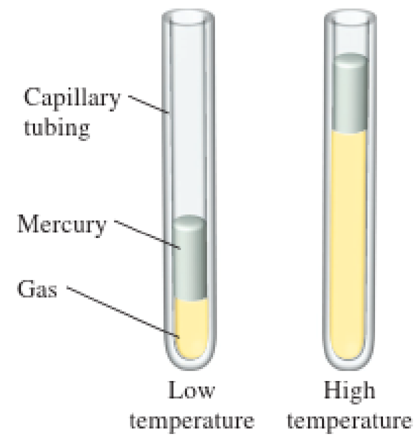


Figure 5.8 *Variation of the volume of a gas sample with temperature, at constant pressure. The pressure exerted on the gas is the sum of the atmospheric pressure and the pressure due to the weight of the mercury.*

[†]Jacques Alexandre Cesar Charles (1746–1823). French physicist. He was a gifted lecturer, an inventor of scientific apparatus, and the first person to use hydrogen to inflate balloons.

[‡]Joseph Louis Gay-Lussac (1778–1850). French chemist and physicist. Like Charles, Gay-Lussac was a balloon enthusiast. Once he ascended to an altitude of 20,000 ft to collect air samples for analysis.

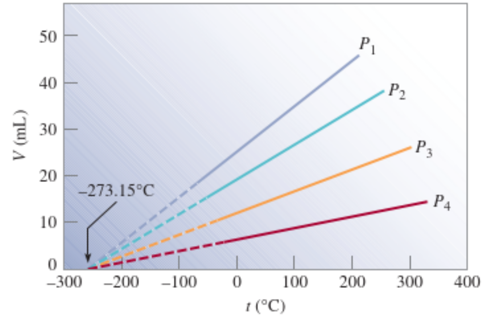


Figure 5.9 *Variation of the volume of a gas sample with temperature, at constant pressure. Each line represents the variation at a certain pressure. The pressures increase from P_1 to P_4. All gases ultimately condense (become liquids) if they are cooled to sufficiently low temperatures; the solid portions of the lines represent the temperature region above the condensation point. When these lines are extrapolated, or extended (the dashed portions), they all intersect at the point representing zero volume and a temperature of −273.15°C.*

In 1848 Lord Kelvin† realized the significance of this phenomenon. He identified −273.15°C as ***absolute zero,*** *theoretically the lowest attainable temperature.* Then he set up an ***absolute temperature scale,*** now called the ***Kelvin temperature scale,*** with *absolute zero as the starting point* (see Section 1.7). On the Kelvin scale, one kelvin (K) is equal *in magnitude* to one degree Celsius. The only difference between the absolute temperature scale and the Celsius scale is that the zero position is shifted. Important points on the two scales match up as follows:

	Kelvin Scale	Celsius Scale
Absolute zero	0 K	−273.15°C
Freezing point of water	273.15 K	0°C
Boiling point of water	373.15 K	100°C

Under special experimental conditions, scientists have succeeded in approaching absolute zero to within a small fraction of a kelvin.

The conversion between °C and K is given in Section 1.7. In most calculations we will use 273 instead of 273.15 as the term relating K and °C. By convention, we use T to denote absolute (kelvin) temperature and t to indicate temperature on the Celsius scale.

The dependence of the volume of a gas on temperature is given by

$$V \propto T$$

$$V = k_2 T$$

or

$$\frac{V}{T} = k_2 \tag{5.3}$$

Remember that temperature must be in kelvins in gas law calculations.

where k_2 is the proportionality constant. Equation (5.3) is known as ***Charles' and Gay-Lussac's law,*** or simply ***Charles' law,*** which states that *the volume of a fixed amount of gas maintained at constant pressure is directly proportional to the absolute temperature of the gas.* Charles' law is also illustrated in Figure 5.6. We see that the proportionality constant k_2 in Equation (5.3) is equal to nR/P.

Just as we did for pressure-volume relationships at constant temperature, we can compare two sets of volume-temperature conditions for a given sample of gas at constant pressure. From Equation (5.3) we can write

$$\frac{V_1}{T_1} = k_2 = \frac{V_2}{T_2}$$

or

$$\boxed{\frac{V_1}{T_1} = \frac{V_2}{T_2}} \tag{5.4}$$

where V_1 and V_2 are the volumes of the gas at temperatures T_1 and T_2 (both in kelvins), respectively.

Another form of Charles' law shows that at constant amount of gas and volume, the pressure of a gas is proportional to temperature; that is

$$P \propto T$$
$$P = k_3 T$$

or

$$\frac{P}{T} = k_3 \tag{5.5}$$

From Figure 5.6 we see that $k_3 = nR/V$. Starting with Equation (5.5), we have

$$\frac{P_1}{T_1} = k_3 = \frac{P_2}{T_2}$$

or

$$\boxed{\frac{P_1}{T_1} = \frac{P_2}{T_2}} \tag{5.6}$$

where P_1 and P_2 are the pressures of the gas at temperatures T_1 and T_2, respectively.

Avogadro's Law: Relating Volume and Amount

In 1808 the French chemist Joseph Louis Gay-Lussac (1778–1850) concluded from experiments on gas reactions that the volumes of reactant gases at the same pressure and temperature are in ratios of small whole numbers (*the law of combining volumes*). For example, two volumes of hydrogen react with one volume of oxygen gas to produce water.

$$\underset{\text{2 volumes}}{2H_2(g)} + \underset{\text{1 volume}}{O_2(g)} \longrightarrow 2H_2O(g)$$

Three years later, the Italian chemist Amedeo Avogadro (1776–1856) interpreted the law of combining volumes in terms of what we now call **Avogadro's law**: *equal volumes of any two gases at the same temperature and pressure contain the same number of molecules.* Thus, two volumes of hydrogen contain twice the number of molecules as in one volume of oxygen, in agreement with the chemical equation for the reaction.

One mole of any gas contains the same number of molecules (Avogadro's number = 6.02×10^{23}) and by Avogadro's law must occupy the same volume at a given temperature and pressure. This *volume of one mole of gas* is called the **molar gas volume, V_m.** Volumes of gases are often compared at **standard temperature and pressure (STP)**, *the reference conditions for gases chosen by convention to be 0°C and 1 atm pressure.* At STP, the molar gas volume is found to be 22.4 L/mol (Figure 5.12).

We can re-express Avogadro's law as follows: the molar gas volume at a given temperature and pressure is a specific constant independent of the nature of the gas.

> Avogadro's law:
> V_m = specific constant (=22.4 L/mol at STP)
> (depending on T and P but independent of the gas)

Table 5.4 lists the molar volumes of several gases. The values agree within about 2% of the value expected from the empirical gas laws (the "ideal gas" value). We will discuss the reason for the deviations from this value at the end of this chapter, in the section on real gases.

Table 5.4 Molar Volumes of Several Gases at 0.0°C and 1.00 atm

Gas	Molar Volume (L)
He	22.40
H_2	22.43
O_2	22.39
CO_2	22.29
NH_3	22.09
Ideal gas*	22.41

*An ideal gas follows the empirical gas laws.

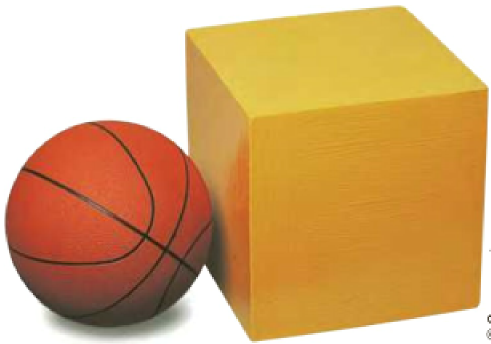


Figure 5.12 ◀

The molar volume of a gas The box has a volume of 22.4 L, equal to the molar volume of a gas at STP. The basketball is shown for comparison.

5.4 The Ideal Gas Equation

Let us summarize the gas laws we have discussed so far:

$$\text{Boyle's law: } V \propto \frac{1}{P} \qquad (\text{at constant } n \text{ and } T)$$

$$\text{Charles' law: } V \propto T \qquad (\text{at constant } n \text{ and } P)$$

$$\text{Avogadro's law: } V \propto n \qquad (\text{at constant } P \text{ and } T)$$

We can combine all three expressions to form a single master equation for the behavior of gases:

$$V \propto \frac{nT}{P}$$

$$V = R\frac{nT}{P}$$

or

$$\boxed{PV = nRT} \tag{5.8}$$

Keep in mind that the ideal gas equation, unlike the gas laws discussed in Section 5.3, applies to systems that do not undergo changes in pressure, volume, temperature, and amount of a gas.

where ***R,*** *the proportionality constant,* is called the ***gas constant.*** Equation (5.8), which is called the ***ideal gas equation,*** *describes the relationship among the four variables P, V, T, and n.* An ***ideal gas*** is *a hypothetical gas whose pressure-volume-temperature behavior can be completely accounted for by the ideal gas equation.* The molecules of an ideal gas do not attract or repel one another, and their volume is negligible compared with the volume of the container. Although there is no such thing in nature as an ideal gas, the ideal gas approximation works rather well for most reasonable temperature and pressure ranges. Thus, we can safely use the ideal gas equation to solve many gas problems.

Before we can apply the ideal gas equation to a real system, we must evaluate the gas constant R. At 0°C (273.15 K) and 1 atm pressure, many real gases behave like an ideal gas. Experiments show that under these conditions, 1 mole of an ideal gas occupies

Figure 5.11 *A comparison of the molar volume at STP (which is approximately 22.4 L) with a basketball.*

22.414 L, which is somewhat greater than the volume of a basketball, as shown in Figure 5.11. *The conditions 0°C and 1 atm* are called ***standard temperature and pressure,*** often abbreviated ***STP.*** From Equation (5.8) we can write

$$\begin{aligned} R &= \frac{PV}{nT} \\ &= \frac{(1\ \text{atm})(22.414\ \text{L})}{(1\ \text{mol})(273.15\ \text{K})} \\ &= 0.082057\ \frac{\text{L} \cdot \text{atm}}{\text{K} \cdot \text{mol}} \\ &= 0.082057\ \text{L} \cdot \text{atm/K} \cdot \text{mol} \end{aligned}$$

The gas constant can be expressed in different units (see Appendix 1).

The dots between L and atm and between K and mol remind us that both L and atm are in the numerator and both K and mol are in the denominator. For most calculations, we will round off the value of R to three significant figures (0.0821 L · atm/K · mol) and use 22.41 L for the molar volume of a gas at STP.

Example 5.3 shows that if we know the quantity, volume, and temperature of a gas, we can calculate its pressure using the ideal gas equation. Unless otherwise stated, we assume that the temperatures given in °C in calculations are exact so that they do not affect the number of significant figures.

Student Hot Spot

Student data indicate you may struggle with the ideal gas law. Access your eBook for additional Learning Resources on this topic.

Example 5.3

Sulfur hexafluoride (SF_6) is a colorless and odorless gas. Due to its lack of chemical reactivity, it is used as an insulator in electronic equipment. Calculate the pressure (in atm) exerted by 1.82 moles of the gas in a steel vessel of volume 5.43 L at 69.5°C.

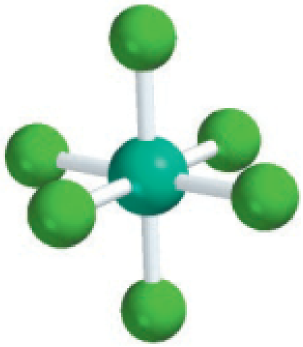
Strategy The problem gives the amount of the gas and its volume and temperature. Is the gas undergoing a change in any of its properties? What equation should we use to solve for the pressure? What temperature unit should we use?

Solution Because no changes in gas properties occur, we can use the ideal gas equation to calculate the pressure. Rearranging Equation (5.8), we write

$$\begin{aligned} P &= \frac{nRT}{V} \\ &= \frac{(1.82 \text{ mol})(0.0821 \text{ L} \cdot \text{atm/K} \cdot \text{mol})(69.5 + 273) \text{ K}}{5.43 \text{ L}} \\ &= 9.42 \text{ atm} \end{aligned}$$

Practice Exercise Calculate the volume (in liters) occupied by 2.12 moles of nitric oxide (NO) at 6.54 atm and 76°C.

Similar problem: 5.32.



SF_6

NH_3

Industrial ammonia refrigeration system.

Courtesy of Industrial Refrigeration Service, Inc. & MTC Logistics

Example 5.4

Ammonia gas is used as a refrigerant in food processing and storage industries. Calculate the volume (in liters) occupied by 7.40 g of NH_3 at STP.

Strategy What is the volume of 1 mole of an ideal gas at STP? How many moles are there in 7.40 g of NH_3?

Solution Recognizing that 1 mole of an ideal gas occupies 22.41 L at STP and using the molar mass of NH_3 (17.03 g), we write the sequence of conversions as

$$\text{grams of } NH_3 \longrightarrow \text{moles of } NH_3 \longrightarrow \text{liters of } NH_3 \text{ at STP}$$

so the volume of NH_3 is given by

$$\begin{aligned} V &= 7.40\ \cancel{\text{g NH}_3} \times \frac{1\ \cancel{\text{mol NH}_3}}{17.03\ \cancel{\text{g NH}_3}} \times \frac{22.41\ \text{L}}{1\ \cancel{\text{mol NH}_3}} \\ &= 9.74\ \text{L} \end{aligned}$$

It is often true in chemistry, particularly in gas law calculations, that a problem can be solved in more than one way. Here the problem can also be solved by first converting 7.40 g of NH_3 to number of moles of NH_3, and then applying the ideal gas equation ($V = nRT/P$). Try it.

Check Because 7.40 g of NH_3 is smaller than its molar mass, its volume at STP should be smaller than 22.41 L. Therefore, the answer is reasonable.

Practice Exercise What is the volume (in liters) occupied by 49.8 g of HCl at STP?

Similar problem: 5.40.

The ideal gas equation is useful for problems that do not involve changes in *P, V, T,* and *n* for a gas sample. Thus, if we know any three of the variables we can calculate the fourth one using the equation. At times, however, we need to deal with changes in pressure, volume, and temperature, or even in the amount of gas. When conditions change, we must employ a modified form of the ideal gas equation that takes into account the initial and final conditions. We derive the modified equation as follows. From Equation (5.8),

The subscripts 1 and 2 denote the initial and final states of the gas, respectively.

$$R = \frac{P_1V_1}{n_1T_1} \text{ (before change)} \quad \text{and} \quad R = \frac{P_2V_2}{n_2T_2} \text{ (after change)}$$

Therefore,

$$\boxed{\frac{P_1V_1}{n_1T_1} = \frac{P_2V_2}{n_2T_2}} \tag{5.9}$$

It is interesting to note that all the gas laws discussed in Section 5.3 can be derived from Equation (5.9). If $n_1 = n_2$, as is usually the case because the amount of gas normally does not change, the equation then becomes

$$\boxed{\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}} \tag{5.10}$$

Applications of Equation (5.9) are shown in Examples 5.5 through 5.7.

Example 5.5

An inflated helium balloon with a volume of 0.55 L at sea level (1.0 atm) is allowed to rise to a height of 6.5 km, where the pressure is about 0.40 atm. Assuming that the temperature remains constant, what is the final volume of the balloon?

Strategy The amount of gas inside the balloon and its temperature remain constant, but both the pressure and the volume change. What gas law do you need?

Solution We start with Equation (5.9)

$$\frac{P_1V_1}{n_1T_1} = \frac{P_2V_2}{n_2T_2}$$

Because $n_1 = n_2$ and $T_1 = T_2$,

$$P_1V_1 = P_2V_2$$

which is Boyle's law [see Equation (5.2)]. The given information is tabulated:

Initial Conditions	Final Conditions
$P_1 = 1.0$ atm	$P_2 = 0.40$ atm
$V_1 = 0.55$ L	$V_2 = ?$

Therefore,

$$\begin{aligned} V_2 &= V_1 \times \frac{P_1}{P_2} \\ &= 0.55\text{ L} \times \frac{1.0\text{ atm}}{0.40\text{ atm}} \\ &= 1.4\text{ L} \end{aligned}$$

Check When pressure applied on the balloon is reduced (at constant temperature), the helium gas expands and the balloon's volume increases. The final volume is greater than the initial volume, so the answer is reasonable.

Practice Exercise A sample of chlorine gas occupies a volume of 946 mL at a pressure of 726 mmHg. Calculate the pressure of the gas (in mmHg) if the volume is reduced at constant temperature to 154 mL.

Similar problem: 5.19.

A scientific research helium balloon.

Source: National Scientific Balloon Facility/ Palestine, Texas

Example 5.6

Argon is an inert gas used in lightbulbs to retard the vaporization of the tungsten filament. A certain lightbulb containing argon at 1.20 atm and 18°C is heated to 85°C at constant volume. Calculate its final pressure (in atm).

Electric lightbulbs are usually filled with argon.

Strategy The temperature and pressure of argon change but the amount and volume of gas remain the same. What equation would you use to solve for the final pressure? What temperature unit should you use?

Solution Because $n_1 = n_2$ and $V_1 = V_2$, Equation (5.9) becomes

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

(Continued)

which is Charles' law [see Equation (5.6)]. Next we write

Remember to convert °C to K when solving gas law problems.

Initial Conditions	Final Conditions
$P_1 = 1.20$ atm	$P_2 = ?$
$T_1 = (18 + 273)$ K $= 291$ K	$T_2 = (85 + 273)$ K $= 358$ K

The final pressure is given by

$$\begin{aligned} P_2 &= P_1 \times \frac{T_2}{T_1} \\ &= 1.20 \text{ atm} \times \frac{358 \text{ K}}{291 \text{ K}} \\ &= 1.48 \text{ atm} \end{aligned}$$

One practical consequence of this relationship is that automobile tire pressures should be checked only when the tires are at normal temperatures. After a long drive (especially in the summer), tires become quite hot, and the air pressure inside them rises.

Check At constant volume, the pressure of a given amount of gas is directly proportional to its absolute temperature. Therefore the increase in pressure is reasonable.

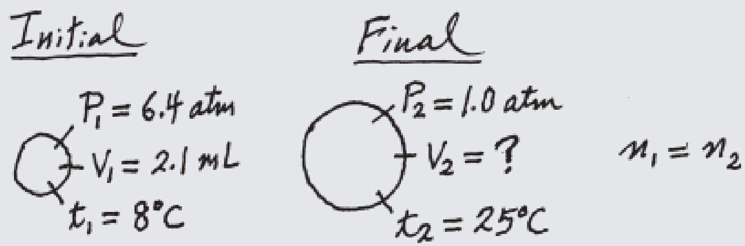
Practice Exercise A sample of oxygen gas initially at 0.97 atm is cooled from 21°C to −68°C at constant volume. What is its final pressure (in atm)?

Similar problem: 5.36.

Example 5.7

A small bubble rises from the bottom of a lake, where the temperature and pressure are 8°C and 6.4 atm, to the water's surface, where the temperature is 25°C and the pressure is 1.0 atm. Calculate the final volume (in mL) of the bubble if its initial volume was 2.1 mL.

Strategy In solving this kind of problem, where a lot of information is given, it is sometimes helpful to make a sketch of the situation, as shown here:



What temperature unit should be used in the calculation?

Solution According to Equation (5.9)

$$\frac{P_1V_1}{n_1T_1} = \frac{P_2V_2}{n_2T_2}$$

We assume that the amount of air in the bubble remains constant, that is, $n_1 = n_2$ so that

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

which is Equation (5.10). The given information is summarized:

We can use any appropriate units for volume (or pressure) as long as we use the same units on both sides of the equation.

Initial Conditions	Final Conditions
$P_1 = 6.4$ atm	$P_2 = 1.0$ atm
$V_1 = 2.1$ mL	$V_2 = ?$
$T_1 = (8 + 273)$ K $= 281$ K	$T_2 = (25 + 273)$ K $= 298$ K

(Continued)

Rearranging Equation (5.10) gives

$$\begin{aligned} V_2 &= V_1 \times \frac{P_1}{P_2} \times \frac{T_2}{T_1} \\ &= 2.1 \text{ mL} \times \frac{6.4 \text{ atm}}{1.0 \text{ atm}} \times \frac{298 \text{ K}}{281 \text{ K}} \\ &= 14 \text{ mL} \end{aligned}$$

Check We see that the final volume involves multiplying the initial volume by a ratio of pressures (P_1/P_2) and a ratio of temperatures (T_2/T_1). Recall that volume is inversely proportional to pressure, and volume is directly proportional to temperature. Because the pressure decreases and temperature increases as the bubble rises, we expect the bubble's volume to increase. In fact, here the change in pressure plays a greater role in the volume change.

Practice Exercise A gas initially at 4.0 L, 1.2 atm, and 66°C undergoes a change so that its final volume and temperature are 1.7 L and 42°C. What is its final pressure? Assume the number of moles remains unchanged.

Similar problem: 5.35.

Example 5.6 Using the Ideal Gas Law

Gaining Mastery Toolbox

Critical Concept 5.6

The ideal gas law ($PV = nRT$) is applied to systems that are not changing. Using the ideal gas law you can determine the value of one of the variables (P, V, n, or T) while the remaining values represented in the equation are known.

Solution Essentials:

- Ideal gas law ($PV = nRT$)
- Units of pressure (Table 5.2)
- Kelvin temperature

Answer the question asked in the chapter opening: how many grams of oxygen, O_2, are there in a 50.0-L gas cylinder at 21°C when the oxygen pressure is 15.7 atm?

Problem Strategy In asking for the mass of oxygen, we are in effect asking for moles of gas, n, because mass and moles are directly related. The problem gives P, V, and T, so you can use the ideal gas law to solve for $\boldsymbol{n}$. The proper value to use for R depends on the units of P and V. Here they are in atmospheres and liters, respectively, so you use $R = 0.0821\ \text{L·atm/(K·mol)}$ from Table 5.5. Once you have solved for n, convert to mass by using the molar mass of oxygen.

Solution The data given in the problem are

Variable	*Value*
P	15.7 atm
V	50.0 L
T	$(21 + 273)\ \text{K} = 294\ \text{K}$
n	?

Solving the ideal gas law for n gives

$$n = \frac{PV}{RT}$$

Substituting the data gives

$$n = \frac{15.7\ \cancel{\text{atm}} \times 50.0\ \cancel{\text{L}}}{0.0821\ \cancel{\text{L}}\cdot\cancel{\text{atm}}/(\cancel{\text{K}}\cdot\text{mol}) \times 294\ \cancel{\text{K}}} = 32.5\ \text{mol}$$

and converting moles to mass of oxygen yields

$$32.5\ \cancel{\text{mol O}_2} \times \frac{32.0\ \text{g O}_2}{1\cancel{\text{mol O}_2}} = \mathbf{1.04 \times 10^3 g\ O_2}$$

5.5 Gas Mixtures; Law of Partial Pressures

While studying the composition of air, John Dalton concluded in 1801 that each gas in a mixture of unreactive gases acts, as far as its pressure is concerned, as though it were the only gas in the mixture. To illustrate, consider two 1-L flasks, one filled with helium to a pressure of 152 mmHg at a given temperature and the other filled with hydrogen to a pressure of 608 mmHg at the same temperature. Suppose all of the helium in the one flask is put in with the hydrogen in the other flask (see Figure 5.19). After the gases are mixed in one flask, each gas occupies a volume of one liter, just as before, and has the same temperature.

According to Dalton, each gas exerts the same pressure it would exert if it were the only gas in the flask. Thus, the pressure exerted by helium in the mixture is 152 mmHg, and the pressure exerted by hydrogen in the mixture is 608 mmHg. The total pressure exerted by the gases in the mixture is 152 mmHg + 608 mmHg = 760 mmHg.

Partial Pressures and Mole Fractions

The pressure exerted by a particular gas in a mixture is the **partial pressure** of that gas. The partial pressure of helium in the preceding mixture is 152 mmHg; the partial pressure of hydrogen in the mixture is 608 mmHg. According to **Dalton's law of**

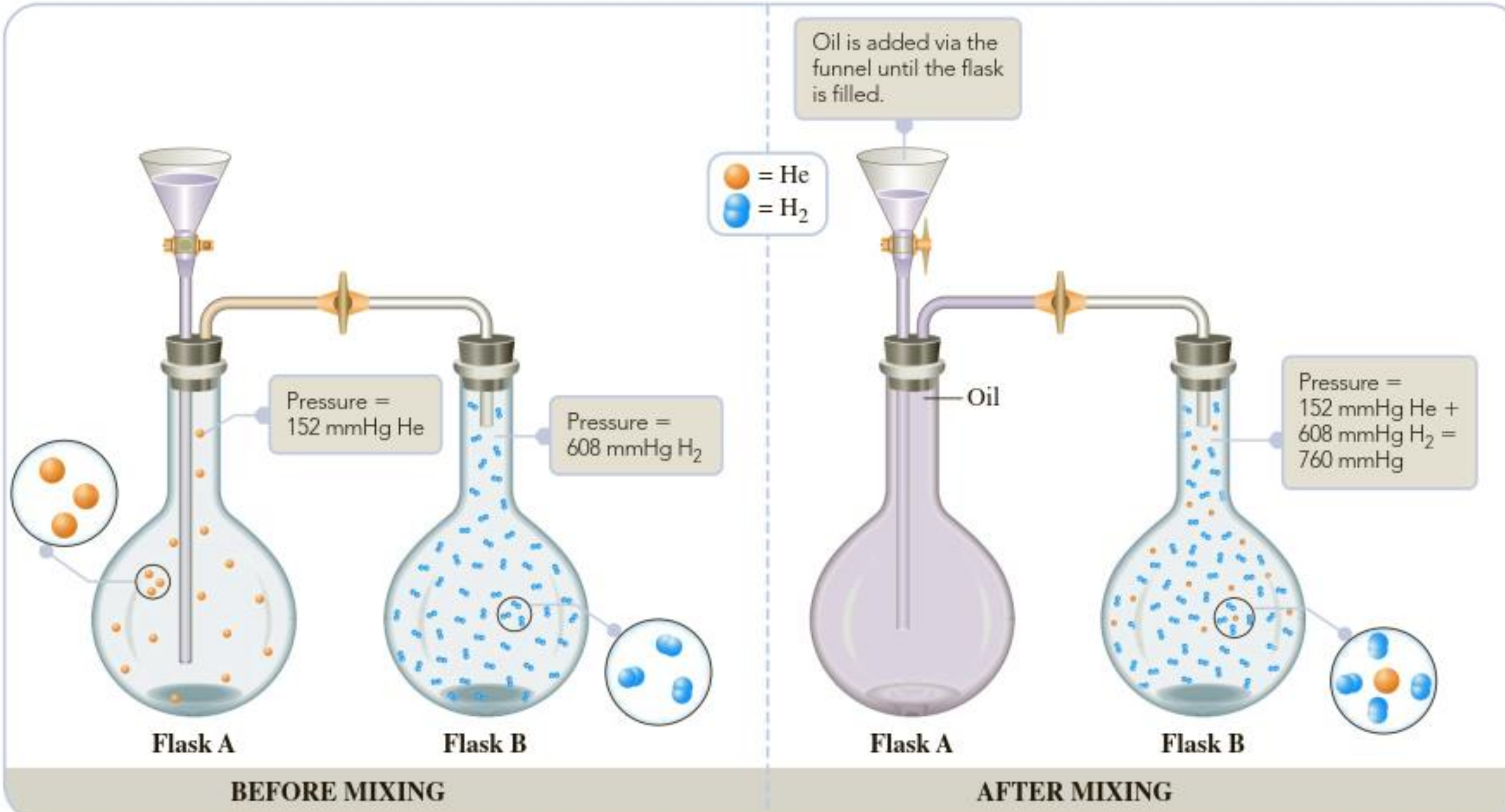


Figure 5.19 ▲

An illustration of Dalton's law of partial pressures The valve connecting the flasks and the valve at the funnel are opened so that flask A fills with mineral oil. The helium flows into flask B (having the same volume as flask A), where it mixes with hydrogen. Each gas exerts the pressure it would exert if the other were not there.

partial pressures, *the sum of the partial pressures of all the different gases in a mixture is equal to the total pressure of the mixture.*

If you let P be the total pressure and P_A, P_B, P_C, . . . be the partial pressures of the component gases in a mixture, the law of partial pressures can be written as

Dalton's law of partial pressures:
$$P = P_A + P_B + P_C + \cdots$$

The individual partial pressures follow the ideal gas law. For component A,

$$P_A V = n_A RT$$

where n_A is the number of moles of component A.

The composition of a gas mixture is often described in terms of the mole fractions of component gases. The **mole fraction** of a component gas is *the fraction of moles of that component in the total moles of gas mixture* (or the fraction of molecules that are component molecules). Because the pressure of a gas is proportional to moles, for fixed volume and temperature ($P = nRT/V \propto n$), the mole fraction also equals the partial pressure divided by total pressure.

$$\text{Mole fraction of } A = \frac{n_A}{n} = \frac{P_A}{P}$$

Mole percent equals mole fraction $\times$ 100. Mole percent is equivalent to the percentage of the molecules that are component molecules.

Example 5.10 Calculating Partial Pressures and Mole Fractions of a Gas in a Mixture

Gaining Mastery Toolbox

Critical Concept 5.10

Each gas in a mixture of gases can be treated independently. This concept is based on ideal gases, with each gas in the mixture having no interaction with the other gases.

Solution Essentials:

- Dalton's law of partial pressure
- Partial pressure
- Mole fraction
- Ideal gas law ($PV = nRT$)
- Units of pressure (Table 5.2)
- Kelvin temperature

A 1.00-L sample of dry air at 25°C and 786 mmHg contains 0.925 g N_2, plus other gases including oxygen, argon, and carbon dioxide.

a. What is the partial pressure (in mmHg) of N_2 in the air sample?

b. What is the mole fraction and mole percent of N_2 in the mixture?

Problem Strategy The key to understanding how to solve this problem is to assume that each gas in the container is independent of other gases; that is, it occupies the entire volume of the sample. Therefore, each gas in the mixture follows the ideal gas law. Using the ideal gas law, we can solve for the pressure of just the N_2, which is its partial pressure. The mole fraction of N_2 can be calculated from the partial pressure of N_2 and the total pressure of all the gases in the sample.

Solution a. Each gas in a mixture follows the ideal gas law. To calculate the partial pressure of N_2, you convert 0.925 g N_2 to moles N_2.

$$0.925\ \cancel{\text{g N}_2} \times \frac{1\ \text{mol N}_2}{28.0\ \cancel{\text{g N}_2}} = 0.0330\ \text{mol N}_2$$

You substitute into the ideal gas law (noting that 25°C is 298 K).

$$\begin{aligned} P_{N_2} &= \frac{n_{N_2}RT}{V} \\ &= \frac{0.0330\ \cancel{\text{mol}} \times 0.0821\ \cancel{\text{L}}\cdot\text{atm}/(\cancel{\text{K}}\cdot\cancel{\text{mol}}) \times 298\ \cancel{\text{K}}}{1.00\ \cancel{\text{L}}} \\ &= 0.807\ \text{atm}\ (\mathbf{= 613\ mmHg}) \end{aligned}$$

b. The mole fraction of N_2 in air is

$$\text{Mole fraction of N}_2 = \frac{P_{N_2}}{P} = \frac{613\ \cancel{\text{mmHg}}}{786\ \cancel{\text{mmHg}}} = \mathbf{0.780}$$

Air contains **78.0 mole percent** of N_2.

5.8 Real Gases

In Section 5.2, we found that, contrary to Boyle's law, the pressure–volume product for O_2 at 0°C was not quite constant, particularly at high pressures (see PV column in Table 5.3). Experiments show that the ideal gas law describes the behavior of real gases quite well at low pressures and moderate temperatures, but not at high pressures and low temperatures.

Figure 5.30 shows the behavior of the pressure–volume product at various pressures for several gases. The gases deviate noticeably from Boyle's law (ideal gas) behavior at high pressures. Also, the deviations differ for each kind of gas. We can explain why this is so by examining the postulates of kinetic theory, from which the ideal gas law can be derived.

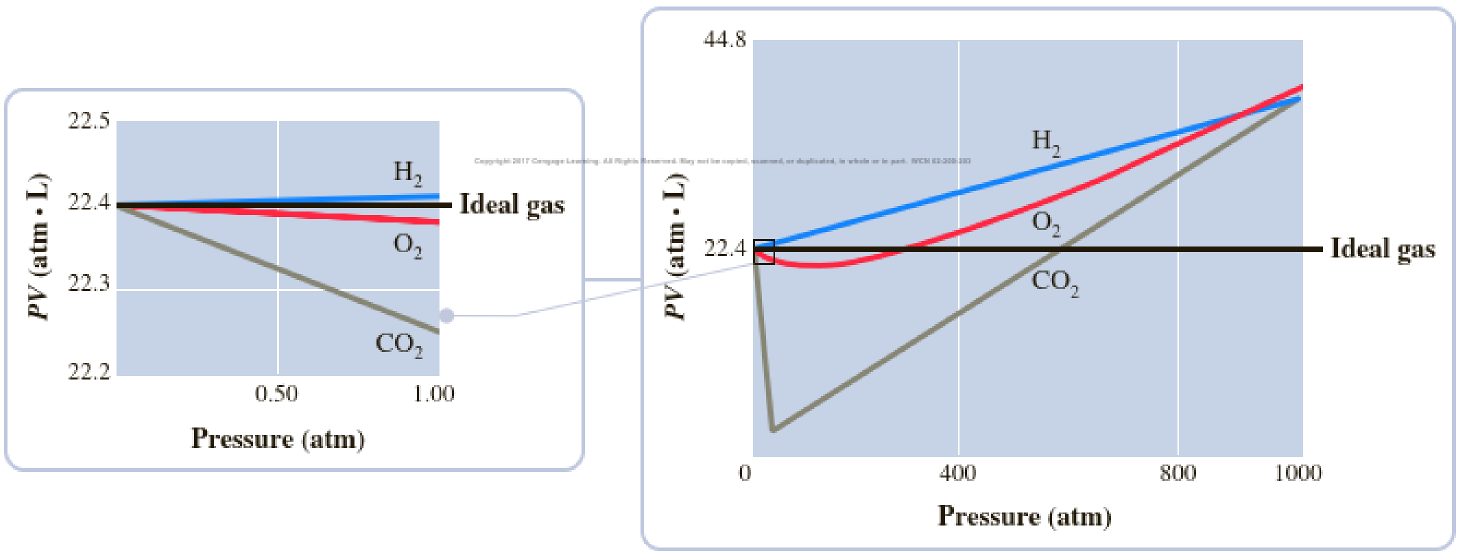


Figure 5.30 ▲

Pressure–volume product of gases at different pressures *Right:* The pressure–volume product of one mole of various gases at 0°C and at different pressures. *Left:* Values at low pressure. For an ideal gas, the pressure–volume product is constant.

a At low pressures, the volume of molecules is a small fraction of the total volume and can be neglected, as in the ideal gas low

b At high pressures, the volume of molecules is a significant fraction of the total volume and cannot be neglected. The ideal gas law is no longer a good approximation.

Figure 5.31

Effect of molecular volume at high pressure

Postulate 1 of kinetic theory says that the volume of space occupied by molecules is negligible compared with the total gas volume. To derive the ideal gas law from kinetic theory, we must assume that each molecule is free to move throughout the entire gas volume, V. At low pressures, where the volume of individual molecules is negligible compared with the total volume available (Figure 5.31a), the ideal gas law is a good approximation. At higher pressures, where the volume of individual molecules becomes important (Figure 5.31b), the space through which a molecule can move is significantly different from V.

Postulate 3 says that the forces of attraction between molecules (intermolecular forces) in a gas are very weak or negligible. This is a good approximation at low pressures, where molecules tend to be far apart, because these forces diminish rapidly as the distance between molecules increases. Intermolecular forces become significant at higher pressures, though, because the molecules tend to be close together. Because of these intermolecular forces, the actual pressure of a gas is less than that predicted by ideal gas behavior. As a molecule begins to collide with a wall surface, neighboring molecules pull this colliding molecule slightly away from the wall, giving a reduced pressure (Figure 5.32).

The Dutch physicist J. D. van der Waals (1837–1923) was the first to account for these deviations of a real gas from ideal gas behavior. The **van der Waals equation** is *an equation similar to the ideal gas law, but includes two constants, a and b, to account for deviations from ideal behavior.*

$$\left(P + \frac{n^2a}{V^2}\right)(V - nb) = nRT$$

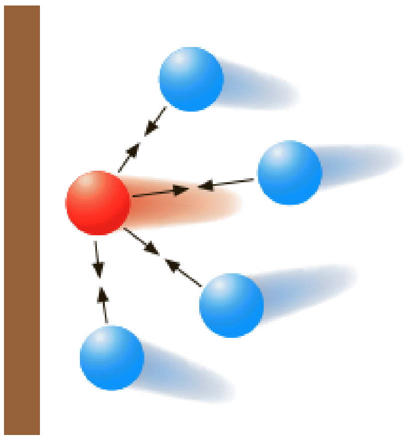


Figure 5.32 ▲

Effect of intermolecular attractions on gas pressure Gas pressure on a wall is due to molecules colliding with it. Here a molecule (shown in red) about to collide with the wall is attracted away from the wall by the weak attractive forces of neighboring molecules (intermolecular forces). As a result, the pressure exerted by the gas is less than that expected in the absence of those forces.

The constants a and b are chosen to fit experiment as closely as possible. Table 5.7 gives values of van der Waals constants for several gases.

You can obtain the van der Waals equation from the ideal gas law, $PV = nRT$, by making the following replacements:

Ideal Gas Equation		*van der Waals Equation*
V	becomes	$V - nb$
P	becomes	$P + n^2a/V^2$

To obtain these substitutions, van der Waals reasoned as follows. The volume available for a molecule to move in equals the gas volume, V, minus the volume occupied by molecules. So he replaced V in the ideal gas law with $V - nb$, where nb represents the volume occupied by n moles of molecules. Then he noted that the total force of attraction on any molecule about to hit a wall is proportional to the concentration of neighboring molecules, n/V. However, the number of molecules about to hit the wall per unit wall area is also proportional to the concentration, n/V. Therefore, the force per unit wall area, or pressure, is reduced from that assumed in the ideal gas law by a factor proportional to n^2/V^2. Letting a be the proportionality constant, we can write

$$P(\text{actual}) = P(\text{ideal}) - n^2a/V^2$$

or

$$P(\text{ideal}) = P(\text{actual}) + n^2a/V^2$$

In other words, you replace P in the ideal gas law by $P + n^2a/V^2$.

The next example illustrates the use of the van der Waals equation to calculate the pressure exerted by a gas under given conditions.

Table 5.7 van der Waals Constants for Some Gases

Gas	a $L^2 \cdot atm/mol^2$	b L/mol
CO_2	3.658	0.04286
C_2H_6	5.570	0.06499
C_2H_5OH	12.56	0.08710
He	0.0346	0.0238
H_2	0.2453	0.02651
O_2	1.382	0.03186
SO_2	6.865	0.05679
H_2O	5.537	0.03049

Example 5.14 Using the van der Waals Equation

Gaining Mastery Toolbox

Critical Concept 5.14
Real gases do not always behave as ideal gases. Gases behave most ideally under low pressure-high temperature conditions. At higher pressures intermolecular forces (attractions between molecules) and the space occupied by the gas particles must be taken into account when making predictions about gas behavior. At low temperatures intermolecular forces must be taken into account.

Solution Essentials:
- Van der Waals equation
  $\left(P + \frac{n^2a}{V^2}\right)(V - nb) = nRT$
- Real gas
- Kinetic-molecular theory
- Ideal gas

If sulfur dioxide were an ideal gas, the pressure at 0.0°C exerted by 1.000 mole occupying 22.41 L would be 1.000 atm (22.41 L is the molar volume of an ideal gas at STP). Use the van der Waals equation to estimate the pressure of this volume of 1.000 mol SO_2 at 0.0°C. See Table 5.7 for values of a and b.

Problem Strategy You first rearrange the van der Waals equation to give P in terms of the other variables.

$$P = \frac{nRT}{V - nb} - \frac{n^2a}{V^2}$$

Now you substitute $R = 0.08206\ L \cdot atm/(K \cdot mol)$, $T = 273.2$ K, $V = 22.41$ L, $a = 6.865\ L^2 \cdot atm/mol^2$, and $b = 0.05679$ L/mol.

Solution

$$P = \frac{1.000\ \cancel{mol} \times 0.08206\ \cancel{L} \cdot atm/(\cancel{K} \cdot \cancel{mol}) \times 273.2\ \cancel{K}}{22.41\ \cancel{L} - (1.000\ \cancel{mol} \times 0.05679\ \cancel{L}/\cancel{mol})} - \frac{(1.000\ \cancel{mol})^2 \times 6.865\ \cancel{L^2} \cdot atm/\cancel{mol^2}}{(22.41\ \cancel{L})^2}$$

$$= 1.003\ atm - 0.014\ atm = \mathbf{0.989\ atm}$$

Note that the first term, $nRT/(V - nb)$, gives the pressure corrected for molecular volume. This pressure, 1.003 atm, is slightly higher than the ideal value, 1.000 atm, because the volume through which a molecule is free to move, $V - nb$, is smaller than what is assumed for an ideal gas (V). Therefore, the concentration of molecules in this volume is greater, and the number of collisions per unit surface area is expected to be greater. However, the last term, n^2a/V^2, corrects for intermolecular forces, which reduce the force of the collisions, and this reduces the pressure to 0.989 atm.